Zhuoren Li /李拙人

- Currently pursuing Ph.D. degree in School of Automobile Studies, Tongji University, Shanghai, China.
- B.S. degree in School of Aerospace Engineering and Applied Mechanics, Tongji University, Shanghai, China.
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Research Experience

- **Interest of Autonomous Driving**: focusing on *Motion Planning, Interaction Decision, Reinforcement Learning, Uncertainty*.
  - Safety enhanced reinforcement learning for behavior decision making.
  - Integrated smoothed motion planning in uncertainty environment based on POMDP.
  - Optimization/Sample-based efficient trajectory planning.
- **Technology Director of 'Tiev-Plus' Electric Autonomous Vehicle** of Institute of Intelligent vehicles.
  - Software system architecture.
  - Planning algorithm integration and Trajectory tracking.
- **Student Director of the Research Group of Decision Making and Trajectory Planning.**
  - Research guidance for 3 PhD students, 7 graduate students.
  - Funding writing assistance and project management.
- **Main Publications:**

[1] **Zhuoren Li**, Jia Hu, Bo Leng, et.al. An Integrated of Decision Making and Motion Planning Framework for Enhanced Oscillation-Free Capability. *IEEE Trans. Intell. Transp. Syst.*, early access, 2023.

[2] **Zhuoren Li**, Lu Xiong, Bo Leng et.al. Safe Reinforcement Learning of Lane Change Decision Making with Risk-Fused Constraint, in *Proc. IEEE Int. Conf. Intell. Transp. Syst. (ITSC)*, 2023, pp. 1313-1319.

[3] **Zhuoren Li**, Lu Xiong Bo Leng. A Unified Trajectory Planning and Tracking Control Framework for Autonomous Overtaking Based on Hierarchical MPC. in *Proc. IEEE Int. Intell. Transp. Syst.*, 2022, pp. 937-944.

[4] **Zhuoren Li**, Bo Leng, Chagen Luo, et.al. A Real-time Path Planner based on Improved Tentacle algorithm for Autonomous Vehicles, in *Proc. IEEE International Conference on Big Data, Artificial Intelligence and Internet of Things Engineering (ICBAIE)*, 2022, pp. 629-634.

[5] **Zhuoren Li**, Lu Xiong, Bo Leng, et.al. Path Planning Method for Perpendicular Parking Based on Vehicle Kinematics Model Using MPC Optimization. *SAE Technical Papers*, 2022-01-0085, 2022.

[6] **Zhuoren Li**, Lu Xiong, Dequan Zeng, et.al. Real-time Local Path Planning for Intelligent Vehicle combining Tentacle Algorithm and B-spline Curve . *IFAC-PapersOnLine*, 2021, 54(10): 51-58.

[7] **Zhuoren Li**, Bo Leng, Jia Hu, et.al. Safety Enhanced Reinforcement Learning for Autonomous Driving: Dare to Make Mistakes to Learn Faster and Better. *IEEE Trans. Intell. Vehicles.*(under review)

[8] Bo Leng, Lu Xiong, **Zhuoren Li***, et.al. Multi-Mode Evasion Assistance Control Method considering Human Driver Operation. *IEEE Trans. Control Syst. Tech.* (under review)

[9] Ruolin Yang, **Zhuoren Li**, Bo Leng, et.al. Convergent harmonious decision--Lane changing in a more traffic friendly way, *IEEE Trans. Intell. Vehicles.* (under review)

- **Award:**
  - 2022 World Artificial Intelligence Conference AI Driving Simulation Competition, Third Prize in the University Challenge Competition.

Main Project Experience

- High-Mobility Motion Planning and Control Research for Chassis-by-wire All-terrain Unmanned Vehicle with Hybrid-steering, **National Natural Science Foundation of China**, 2024, Project Core Participant.
- The Key Technology of Collaborative Perception and Control of Intelligent Bus System, **National Key Research and Development Program of China**, 2023, Project Core Participant.
- Real-time Multiple Strongly Coupled High-level Autonomous Driving Simulation Software Development, **National Key Research and Development Program of China**, 2022, Subject Participant.
- Binary mixed traffic behavior characteristics and collaborative paradigm, **Science and Technology Commission of Shanghai**, 2022, Subject Participant.
- Development of Advanced Assistance Algorithm for Emergency Collision Avoidance based on Steer-by-Wire System, **Shanghai Automotive Industry Science and Technology Development Foundation**, 2022, Project Technical Director.
- Autonomous Valet Parking (AVP) System Development, Project for Enterprise, 2021, Project Technical Director.
- Development and Application of Automatic Valet Parking System, Project for Enterprise, 2020, Project Technical Director.